

**CONICS CALCULATOR (ConCalc) AS EDUCATIONAL WEBSITE
FOR STEM LEARNERS**

A Research Paper

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Chapter 1

THE PROBLEM AND ITS BACKGROUND

Introduction

The world today is characterized by constant flow of information and a rapid change. Recently, as the world has progressed, the Fourth Industrial Revolution, also known as Industry 4.0, continues to make an impact on various areas, including education (Abersek & Flogie, 2017), with significant research focusing on the use of digital technology. In recent years, the use of artificial intelligence or AI is receiving an increasing amount of attention. This rise of AI is significantly influencing how to learn things, with tools like chatbots, AI generators, and problem-solving application that has becoming popular among students, especially in mathematics education. In the sector of Mathematics, it usually involves using symbols to represent concepts like numbers, quantity, space, and structure which are seen as a complex subject.

In the study of Novriani and Surya (2017), it indicates that students often find solving mathematics problems challenging therefore, many researchers have made attempts to develop various learning strategies and tools to enhance students' mathematics learning outcome. Like artificial intelligence, Gao (2020) stated that as computer technology advances, AI keeps growing and improving. Consequently, allowing students to improve their critical thinking skills with the help of AI. Following this, Popenici and Kerr (2017) also mentioned that technology in higher education is meant to boost human thinking and improve the learning process which AI makes it faster and easier to grasp for students to find

answers. However, there are many platforms like Mathway, Photomath, Wolfram Alpha, and Gauthmath that provide quick answers yet they often lack depth and transparency. Additionally, these platforms often require paid subscriptions or account creation, limiting accessibility, and rarely incorporate visual representations like graphs.

Thus, this study aimed to provide solutions and develop a website that would offer detailed step-by-step instructions to help the Science, Technology, Engineering, and Mathematics (STEM) learners in dealing with the conic sections. The researchers wished to equip students with the abilities needed to solve a variety of equations effectively by providing this website, Conics Calculator (ConCalc). It is a website that was designed to provide a comprehensive information about conic sections equations. It offers direct solutions, explanations of the underlying mathematical principles behind the formulas, and detailed information about each equation that will help the students comprehend and resolve mathematical issues in converting conic sections such as transforming circle from $x^2 + y^2 + Dx + Ey + F = 0$ to $\{x^2 + y^2 = r^2, (x - h)^2 + (y - k)^2 = r^2\}$, converting ellipse from $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$ to $\{x^2/a^2 + y^2/b^2 = 1, x^2/b^2 + y^2/a^2 = 1, (x - h)^2/a^2 + (y - k)^2/b^2 = 1, (x - h)^2/b^2 + (y - k)^2/a^2 = 1\}$, turning the parabola from $Ax^2 + Dx + Ey + F = 0$ and $Cy^2 + Dx + Ey + F = 0$ to $\{x^2 = 4py, x^2 = -4py, y^2 = 4px, y^2 = -4px\}$ and $\{(x - h)^2 = 4p(y - k), (x - h)^2 = -4p(y - k), (y - k)^2 = 4p(x - h), (y - k)^2 = -4p(x - h)\}$, and changing hyperbola from $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$ to $\{x^2/a^2 - y^2/b^2 = 1, y^2/a^2 - x^2/b^2 = 1, (x - h)^2/a^2 - (y - k)^2 = 1, (y -$

$k^2/a^2 - (x - h)^2 = 1\}$. Also, the definition and parts of the conic sections can be easily accessed by students using this innovative intelligence software.

The goal of this study is to shed light on the students' problem in dealing conic sections particularly to conversion of forms, reducing the negative impact on student learning and potential solutions for academic honesty. Additionally, instead of promoting cheating and laziness, the website aimed to deliver personalized learning resources. By tailoring content to individuals' need and offering accurate evaluations, it would likely enhance student engagement, comprehension, and overall academic success.

Background of the Study

Throughout history, scholars from all over the world have significantly contributed to the development of mathematics, a crucial field for abilities like pattern recognition, data analysis, and evidence search. Nowadays, dealing mathematics is such a challenging task for learners, especially Filipino learners. Gafoor et al. (2015) identify two main reasons for learners' difficulties in mathematics: 1) problems with understanding the subject; and 2) instructor-related factors where teachers struggle to explain lessons effectively. Moreover, referring to DepEd Order No. 20 s. 2016 "Guidelines on K-12 Basic Education Program" which was implemented by the Department of Education (DepEd) last May 2016, which stated that STEM learners must take Pre-Calculus as a specialized subject to enhance their skills in applying its concepts in linear equations and other mathematical contents such as conic sections.

DepEd Tambayan (n.d.) mentions the hidden struggles of students which are understanding and addressing learning difficulties. This article discussed how students with learning difficulties often struggle to keep up with the peers, academically leading to frustration, low self-esteem, and lack of confidence with their abilities. However, e-learning is the most popular learning tool among students who can readily access it through gadgets, as the modern world is driven by technological advancements and artificial intelligence (AI). Mensah and Nabie (2021) found that using PowerPoint presentations as an alternative teaching resource helps students stay more focused and motivated. Using an ICT integration approach to teach and learn mathematics concepts has also been shown to increase student achievement in the subject.

The conic sections in Mathematics and Physics offer practical applications for learners, focusing on manipulating and understanding equations and word problems. Moreover, Grade 11 learners often face misconceptions and mistakes in understanding the concept of conic section, including difficulty of identifying simple equations, describing basic parabola equations, and comprehending conic equations (Sudihartinih & Purniati, 2022). Due to this problem, the researchers aimed to create a website that can provide a step-by-step procedure on how to convert one form of equation to another and vice versa of conic sections. The purpose of this study is to guide and help the STEM learners to aid the common mistakes in converting the forms of conic sections.

JavaScript was chosen as a programming language used in developing a teaching platform in hard-to-learn topics in mathematics. As stated, it is the best

choice in presenting information, instructions, and graphs where the learners can understand and focus. With these, the researchers aimed to test the level of acceptability and the level of performance of the Conics Calculator (ConCalc) as an e-learning assistant for learners on how to convert one form of equation to another and vice versa of conic sections.

Theoretical Framework

A. JavaScript Theory

According to Jordana (2024), HTML and CSS are used in web development, with an emphasis on how they work together to enhance user experience. It looks at the concepts, procedures, and ideals associated with these technologies. Using HTML, CSS, and JavaScript, the work involves building webpages. The content is divided into modules by the new HTML structure, and CSS is used for styling. The new standard, CSS3, includes distinct modules for various style types in addition to structural elements. Most work is done through browsers, and candidates can plan meetings to get approved more quickly.

Over the past ten years, there has been a significant increase in the amount of JavaScript use in web pages, resulting in large, complex pages that require a lot of processing power on handheld mobile devices. Because more people are using these devices to access the internet these days and since many mobile web users only use low-end devices, these devices are significant to meet their needs (Chaqfeh et al., 2022).

B. E-learning Theory

As part of the grand theory of connectivism, according to He (2023), the E-learning Theory focuses on how technologies can be used and designed to create new learning opportunities and to promote effective learning. This instructional technology should be designed to minimize unnecessary cognitive load and manage actual and relevant load at levels appropriate for the students. It has been observed that, in comparison to students exposed to text alone, there is a significant increase in student engagement and the development of active learning principles. These principles have also been successfully applied to non-technical learning content and learners of the various age groups.

Conceptual Framework

The diagram depicts the conceptual framework of the variables in Conics Calculator (ConCalc) as Educational Website for STEM Learners. It includes the input, process, and output that are represented by boxes and arrows. The first box which contains HTML, CSS, and JAVASCRIPT represents the starting point of the input. The HTML is the standard markup language for documents designed to be displayed in a web browser. The CSS on the other hand, is a style sheet language used for specifying the presentation and styling of a document written in a markup language such as HTML or XML. Meanwhile, the JavaScript is a programming language and core technology of the Web, alongside HTML and CSS. The second box contains the process of developing the website which includes development or coding, designing, testing, and deployment of the website. Finally, the third box contains the output of the study

which is the Conics Calculator (ConCalc). This website provides step-by-step procedures on how to convert one form of equation to another of conic sections.

Research Paradigm

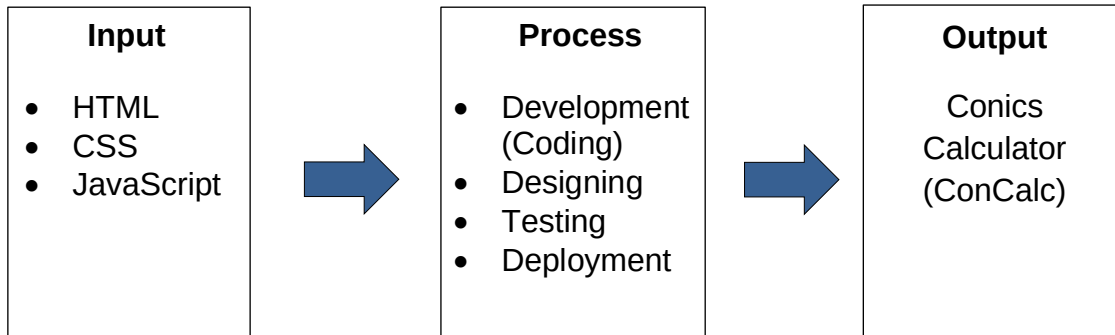


Figure 1

Research Paradigm

Statement of the Problem

This study aimed to determine the level of effectiveness and the level of performance of Conics Calculator (ConCalc) as Educational Website for STEM Learners.

Specifically, it would like to answer the following questions:

1. What were the procedures and algorithms used in developing Conics Calculator (ConCalc) as Educational Website for STEM Learners?
2. What is the level of acceptability of ConCalc with respect to:
 - 2.1 usability; and
 - 2.2 web design?
3. What is the level of performance of ConCalc with respect to:
 - 3.1 accuracy; and
 - 3.2 accessibility?

4. Is there any significant relationship between the level of acceptability and the level of performance of ConCalc?

Hypothesis

H_0 : There is no significant relationship between the level of acceptability and the level of performance of ConCalc.

Significance of the Study

The researchers developed an educational website to enhance the learning in mathematics specifically in the field of Pre-Calculus. Furthermore, the results of the study will benefit the following:

Students. This will help students to better understand solving conic sections specifically on how to convert one form of equation to another. It will also make learning more enjoyable because the ConCalc provides step-by-step procedures.

Teachers. This study will benefit the teachers who deal with the topic of conic sections. This can serve as a teaching material for teachers to use on teaching their students about conic sections.

Future Researchers. This study will benefit them by providing additional information and/or a reference for their own study.

Web Developer. This study provides valuable insights for web developers by focusing on writing and implementing the codes that are essential in building instructional websites, especially those for ConCalc. The codes form the basis for assessing the functionality of the website.

Scope and Delimitation

The researchers aimed to develop a website called Conics Calculator (ConCalc) which resolves and provides step-by-step procedures in converting various forms of conic sections. In addition, the website includes the definition and parts of each conic sections and graph. The researchers used survey questionnaire regarding Conics Calculator (ConCalc) to test its acceptability and performance as educational website for Science, Technology, Engineering, and Mathematics (STEM) learners. The website included functions such as converting forms, giving its coordinates, and graph. The learning aid was available on all devices such as cellphones, laptops, or iPads and etc. However, this study is still progressing on different parts of the conic sections. It was conducted and only limited on STEM learners in Sta. Maria Integrated High School, Academic Year 2024-2025.

Definition of Terms

The terms stated below aimed to have a better understanding of the study. These terms are conceptually and operationally defined based on their application in the study.

Artificial Intelligence (AI). It is a combination of technologies that allows computers to perform a wide range of advanced operations, such as seeing, understanding, and translating spoken and written language, analyzing data, making suggestions, and more.

Cascading Style Sheets. It is a stylesheet language used to determine the display and styling of a document authored in a markup language like HTML or XML.

Circle. It is formed when a plane cuts the cone at right angles to its axis.

Conics Calculator (ConCalc). It is the website developed by the researchers that aims to provide step-by-step instructions in converting one form to another and vice versa of conic sections.

Educational Website. It is an online platform that gives parents, teachers, students, and aspiring learners access to educational materials and tools.

E-learning. It refers to modern learning style with the use of electronic media. It also known as electronic learning.

Ellipse. It is formed when the tilted plane intersects to nappe of the cone.

Graph. It is a visual representation of the conic sections.

Hyperbola. It is a curve that occurs at the intersection of a cone and a plane.

Hypertext Markup Language. It is the standard markup language for documents designed to be displayed in a web browser. It defines the content and structure of web content.

JavaScript. It is often abbreviated as JS. It is a programming language and core technology of the Web, alongside HTML and CSS.

Parabola. It is formed when a plane cuts one nappe of the double-napped cone parallel to the edge of the cone.

Pre-Calculus. It is a subject that act as a pre-requisite for Calculus and cover advanced mathematical concepts based on quantitative reasoning and functions.

Website. It is a collection of files accessed through a web address, covering a particular theme or subject, and managed by a particular person or organization.

Web Development. It refers to the overall process of creating website application.

Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents a comprehensive review of the related literature and studies which are presented in the following paragraphs. The review includes local and international studies, ensuring a balanced exploration of the subject matter while demonstrating its relevance to the research.

Related Literature

A. Web Development

According to the journal of Decker (2024), it is hardly surprising that the web development industry is growing quickly given the rapidly rising number of Internet users. Web developers are expected to see employment growth of 13% between now and 2030, which is substantially faster than most other technology careers. In general, the activities and procedures involved in building and managing a website are referred to as website development. This covers everything from coding and markup to network configuration, scripting, and CMS development.

Based on the journal of Joshi (2023), it can be sure that JavaScript is responsible for any components you encounter on websites that make use of animations, maps, graphics, etc. Most contemporary websites on the internet are built using this object-oriented, lightweight programming language. All of the main web browsers support it and it is compatible with multiple platforms. Websites can be developed with JavaScript on both the front and back ends. When done correctly, JavaScript captures smooth communication between the client and the

server. JavaScript's primary advantage for use in contemporary web development is its ability to manipulate the Document Object Model.

Using different programming languages, like Java, Python, and JavaScript, web development entails building and managing websites and web applications. Developers utilize these programming languages to create the software that drives a website's functionality and interactivity, enabling desktop and mobile device accessibility. In order to provide users with a flawless online experience, web development includes everything from database administration and coding to user interface design. The following tutorials offer pointers and advice on how to begin developing websites (Anonymous, 2024).

B. Educational Website

According to Rafath (2023), educational websites are vital resources that enhance students' educational experiences. Information access, extra resources, individualized and interactive learning, ease of use and accessibility, cooperation, and keeping students abreast of current events are all made possible by them. Students can broaden their perspectives, gain a deeper understanding, and acquire critical skills necessary for success in both their academic and professional endeavors by utilizing instructional websites.

C. Technology

According to Mangal and Mangal (2019), all fields of knowledge now use technology, and education is no exception. The work covers a wide range of topics in educational technology but focuses on contemporary developments and trends in educational technology to give pre-service and in-service teachers the

expertise they need in this field. The book emphasizes the importance of education and numerous other teaching and learning strategies while also covering the objectives, ideas, application, and management of educational technology.

Instructors have noticed that when students use technology in the classroom, they are more likely to assist one another. Many technology-based assignments require additional components, which puts students in situations where they must ask for assistance from their teachers or peers. Additionally, when students work in small groups, technically proficient students can mentor their less familiar with technology peers (Anonymous, 2024).

D. Teaching Material

Conforming to the journal review written by Malalina et al. (2024), the development of mathematics teaching materials for junior high schools, focusing on content, construct, and language. Field tests show these materials can enhance students' analytical and critical-thinking abilities in a maritime context.

In a literature review analyzed by Suripah (2021), fifteen students of class VII SMP Nurul Farah Pekanbaru who were selected by simple random technique were surveyed in using website as a mathematics learning media during online learning period. This shows that the effectiveness of using the website was in the quite good category, 40% of students said the website was very effective, 53.3% said it was effective, and only 6.7% said it was ineffective. Based on this, the website becomes an effective learning medium to be used during the online learning period.

E. Artificial Intelligence (AI)

Emerging technologies such as AlphaGeometry indicate that machines with increasingly human-like reasoning abilities are becoming more and more common. As a result, it might be able to develop more potent artificial intelligence (AI) tools that assist mathematicians in solving equations and possibly even produce improved tutoring resources (Heikkila, 2024).

Based on the investigation of Slimi (2022), with the ability of AI to automate administrative tasks, provide timely feedback, and tailor teaching strategies to each student's needs, AI has the potential to completely transform education. This recommend that higher education programs incorporate AI more thoroughly and institutions take into account the ethical considerations when developing and implementing AI programs.

Related Studies

A. Artificial Intelligence (AI)

Tarek Ait Baha's (2024) study states that artificial intelligence (AI) is becoming more and more prevalent in education. He compared two experimental groups – one using chatbots and the other a traditional method – and found that chatbots improve student learning by reducing stress and saving time. He also found that integrating ai into the classroom encourages better student interaction and proper engagement, both of which improve academic outcomes.

Based on the study of Oposemowa and Adewiyu (2024), the review identifies the research techniques, research instruments, participants, and AI tools utilized in mathematics education by a thorough analysis of ten pertinent

papers from 2015 to 2023 in Scopus and Google Scholar. The use of AI-driven individualized learning and improved instruction in mathematics, real-time evaluation and feedback, curriculum building, and instructor empowerment are some important concepts that were emphasized. This analysis also addresses how AI can be used to solve educational inequities and better prepare students for the demands of 4IR.

For many students, learning mathematics has been a significant problem. The development of computer technology, particularly AI, offers a way to address this issue by identifying the unique learning challenges of each student and offering tailored support to help them achieve their best learning outcomes in math classes (Hwang & Tu, 2021).

The study of Richter et al. (2019) states that applications of AI in education are becoming more common and have drawn a lot of attention in recent years. With a two- to three-year adoption period, AI and adaptive learning technologies are highlighted as significant advancements in educational technology in the 2018 Horizon study. The research states that experts project a 43% increase in AI use in education between 2018 and 2022.

B. Technology

According to Anonymous (2022), the internet has completely changed how people obtain information. These days, everything can learn through the use of technology, practically any issue by just entering a few phrases into a search engine. The education system has been significantly impacted by this ease of

access to information. Students may now learn everything they want by the help of internet.

These days, technology has drastically altered the educational scene. It has an impact on how individuals work, play, live, and learn. Technology has put teaching techniques and strategies—particularly in mathematics—as well as the management of education under pressure (Rizada & Rey, 2023).

Espenosa et al. (2023) highlight the challenges of integrating technology in developing nations like the Philippines, despite its potential to improve education and governance. They emphasize the need to address ongoing issues within the school system, recognizing technology's purpose in addressing problems.

C. Web Development

The study of Jain et al. (2024) looks at how HTML and CSS are used in web development, with an emphasis on how they work together to enhance user experience. It looks at the concepts, procedures, and ideals associated with these technologies. Using HTML, CSS, and JavaScript, the work involves building webpages. The content is divided into modules by the new HTML structure, and CSS is used for styling. The new standard, CSS3, includes distinct modules for various style types in addition to structural elements. Most work is done through browsers, and candidates can plan meetings to get approved more quickly.

D. Website

The goal of the study on the advantages and features of using websites for math instruction is to support students and teachers who may not yet be aware of the value of these resources. As a result, teachers are impacted by technology since it is crucial for students to learn from both books and electronic resources (Sabirova et al., 2019).

In a study conducted by Vina et al. (2022) entitled *Influence of Website-Based E-Learning in the Pandemic Era: Improving Students' Creative Thinking Skills through Educational Management*, the findings showed the impact of the vinnarrai.com website on a fluid. It revealed that students' scores for creative thinking were higher in an experiment class that used the website than in a controlled class that used a textbook. Thus, it appears that in the pandemic era, employing "website-based e-learning" enhanced the students' capacity for creative thought.

Shana et al. (2023) stated that one of the most essential tools for learning and enhancing academic achievement for pupils is educational websites. Students use them to obtain the necessary information, such as homework assignments, learning resources, and assessment data.

As more advanced technologies and suitable applications (such as websites focused around mathematics) have become available, the use of technology to support education has been growing. It is vital to come up with fresh ideas for ways to help learning, but it is also critical to assess current tools

and websites to make sure that they (like websites focused on mathematics) are useful, easy to use, and practical (Ziegenfuss & Smith, 2015).

Based on Williams (2020), students frequently use digital tools and websites like Slader, Photomath, and Wolfram Alpha to accomplish their homework. These applications allow students to input their difficulties and receive step-by-step solutions. Students can either copy the solutions straight from these websites or utilize them to learn the procedure.

According to the study of Serevina (2022) in utilizing a website as opposed to a controlled classroom with a textbook, it appears that, in the epidemic era, employing website-based e-learning has enhanced pupils' capacity for creative thought.

Based on the study of Mthethwa (2015), analysis has been done on the data collection results. In the KwaZulu Natal district of UMkhanyakude, it has been discovered that the use of GeoGebra in Euclidean geometry (circle geometry) instruction improves student performance, comprehension, and justification of circle geometry proofs and theorems.

According Ogbonnaya (2019), conducting qualitative research to find out how students feel about using GeoGebra and other software to study math. The studies should evaluate the attitudes and perceptions of educators regarding the use of ICTs and their integration into mathematics instruction.

The study of Obina et al. (2022) found out that using math apps aids the respondents' academic progress. The fact that the apps improve their academic performance was another indication of their effectiveness. Their grades stabilized

and improved with the aid of the aforementioned applications. There is a noteworthy correlation between the academic performance of the respondents and their study habits. Because of this, using math apps offers a chance to expand the respondents' ongoing learning.

E. Calculator

According to the study of Nguyen (2023), Vietnam has seen a significant increase in the use of calculators in mathematics instruction by both instructors and pupils, particularly with the introduction of the 2006 general education curriculum. A conceptual approach to problem resolution and knowledge acquisition can both benefit from calculators. Innovating teaching and learning processes using calculators, instructors' opinions, and attitudes toward the use of ICT in the classroom are positively impacted.

F. E-learning

Conforming to Fedorowa et al. (2019), digital learning materials are becoming more and more abundant every day. Teachers do not know yet how beneficial these materials are, despite the fact that they can be utilized to teach kids. It is crucial for teachers to be aware of the educational tasks that can be completed with the aid of electronic resources, the methodological roles they play, and the kinds of training activities they support and initiate to incorporate websites into the educational process in a purposeful and philosophically sound manner.

In the study conducted by Bardelle (2013), students were given access to online tools in addition to traditional materials and resources to help them

achieve learning strategies that were primarily focused on the semiotic and pragmatic aspects of mathematical language. From a cognitive and metacognitive standpoint, the online exercises are a useful tool for encouraging students to move from passive learning—which involves listening to lectures and taking notes—to more active learning modalities, such as student-instructor or student-computer interaction exercises.

The study written by Guerrero et al. (2020) demonstrated the beneficial effects of using the e-learning approach on participation, motivation, autonomy, and mathematical concepts as well as grades and results. Given a comparison between the e-learning approach and the expository method, it can be concluded that adult students studying mathematics in the high school educational stage benefit from it. As such, this approach is thought to work well when applied to adults.

Mothila (2015) asserts that e-learning is quickly becoming a more well-liked teaching method in higher education settings because of the extensive use of online systems in education. To improve and enhance teaching and learning, e-learning makes use of information and communication technology (ICT). According to the findings, ICT greatly raises students' academic achievement when they use e-learning. The findings demonstrate that students' overall academic performance in the classroom is greatly enhanced by ICT.

According to the study of Sumarwati et al. (2020) entitled *Developing Mathematics Learning Media Based on ELearning Using Moodle on Geometry Subject to Improve Students' Higher Order Thinking Skills*, it demonstrated that

MOODLE e-learning resources for mathematics effectively motivate students to understand, interpret, evaluate, and manipulate data, especially in geometry. They also encourage critical thinking, new ideas, attention, and active participation in class activities.

The study of Santos (2020) highlighted the demands and actions that arise from educators' utterances when they turn to their own experiences of elaboration when teaching mathematics online. The inquiry uncovers invention, intropathy, and reconfigurations in the teaching of mathematics. Because online resources and venues are connected to new understandings and subject practice, curriculum revisions, technique modifications, and adjustments are necessary for dynamic knowledge production.

The study of Almanthari et al. (2020) revealed that the barrier at the student level had the biggest effect on the use of e-learning. Furthermore, there was a significant positive correlation observed between the barriers at the student, school, and curriculum levels. The study demonstrated that the degree of barriers was unaffected by the backgrounds of the teachers. By emphasizing the value of student voices, this study encourages more conversation about how to maximize the advantages of e-learning both during and after the pandemic. It also helps to overcome obstacles to e-learning.

Khan et al.'s (2020) study revealed the students' positive perception towards e-learning and thus acceptance of this new learning system. It has also empirically demonstrated the significance of e-learning in the time of COVID-19 crisis. In fact, e-learning has emerged as a new way of enhancing the learning

process where social media may further improve the learning output. The findings of the study will facilitate educational institutions and policy makers to take this online-learning process to the next level in a better way.

According to the journal of Loh et al. (2016), the findings showed that although students saw improved learning outcomes and flexibility as benefits of online learning, they also expressed concerns about self-paced learning, lack of human interaction, and the inability to foster teamwork.

To support this study, based on Mammattah (2016), most students believed that e-learning is a novel concept that should be promoted. Nevertheless, a few worries, like the worry that employers will discriminate against students who use e-learning, were found. The preference of the respondents for learning was also discovered to be hybrid learning, which combines online and in-person instruction.

In line with the study of Amponsah et al. (2022), having self-learning skills and owning a computer at home with internet access are equally important for students' academic achievement. Depending on their accessibility and use by senior high school students in Nigeria, it was discovered that the majority of learners felt the internet was far better and more accessible than their school libraries. The study indicates that students view the internet as a broad knowledge resource. It helps them build good study habits, which improves their academic success.

Chapter 3

RESEARCH METHODOLOGY

This chapter contains the locale of the study, research design, subject of the study, sampling technique, research materials, budgetary requirements, techniques and procedures, research instrumentation, data gathering procedures, and statistical data. The methods and procedures are carefully chosen to ensure the validity and reliability, offering a systematic approach to understanding the research problem.

Locale of the Study

The study was conducted in Sta. Maria Integrated High School during the First Semester of the Academic Year 2024-2025. Grade 11 students were specifically chosen since they were taking the Pre-Calculus as a specialized subject.

Research Design

This research study used quasi-experimental research design to determine the level of acceptability and the level of performance of the Conics Calculator (ConCalc) to Grade 11 STEM learners.

Quasi-experimental research is defined as an investigation that resembles experimentation but is not a real experiment. Establishing a cause-and-effect relationship between an independent and dependent variable is the goal of a quasi-experimental design.

Subject of the Study

The respondents in this study were Grade 11 STEM students enrolled at Sta. Maria Integrated High School for the Academic Year 2024-2025. They were taking Preparatory Calculus (Pre-Calculus), a specialized STEM subject that focuses on understanding how things change over time. The course aims to prepare students for dynamic concepts by helping them link their prior knowledge of Algebra and Geometry.

Sampling Technique

The research employed a purposive sampling technique, which was deemed the most suitable for this study. This method is used when the sample needs to possess specific characteristics, meaning only those who meet the criteria are selected. Additionally, the researchers focused on Grade 11 STEM students at Sta. Maria Integrated High School. This focus ensures that the sample aligns closely with the research objectives, as these students are engaged in mathematical conversion and are well-positioned to benefit from using ConCalc as an e-learning tool.

Research Materials

The tools and equipment utilized in the development of Conics Calculator (ConCalc) are given below:

Quantity	Unit	Material's Description
5	pieces	Mobile phone
1	piece	Laptop with visual studio Code Internet

Budgetary Requirements

In this part, the calculated material amount and cost were listed.

Particulars	Amount (Php)
Internet connection	0
Electricity	0
Total	0

Techniques and Procedures

The processes of developing the website were conducted in an orderly manner, and the activities involved in these areas were as followed:

A. Idea Generation and Illustration

First, unlike other advanced conic section calculators, the researchers explained how the website would work and how it would assist STEM learners in understanding how to convert forms of conic sections without just providing the answer right away.

B. Design and Development

While developing the website, the researchers planned and discussed over the essential components that would distinguish it from other websites of a similar nature. The researchers employed JavaScript as their primary programming language after deciding which language was convenient to learn and use.

C. Quality Guarantee and Deployment

Before launching the prototype, the researchers thoroughly tested the website to ensure its functionality and minimize errors. When the

researchers were contented with the website's functionality, the website link was then distributed to the chosen respondents for product testing.

Research Instrumentation

In this research study, the researchers used a survey questionnaire to collect, analyze, and measure data regarding the conic section to determine the level of acceptability and the level of performance of ConCalc. The instrument of the study was inspected and assessed by the researchers to make sure it was accurate and effective before it is distributed. The questionnaire consisted of questions about the usability, web design, accuracy, and accessibility.

The statements in the questionnaire were evaluated based on the numeric weight corresponding with its verbal interpretation as described below:

I. Level of Acceptability with respect to Usability and Web Design

Numeric Weight	Verbal Interpretation
3.26 - 4.00	Highly Acceptable
2.51 - 3.25	Acceptable
1.76 - 2.50	Slightly Acceptable
1.00 - 1.75	Unacceptable

II. Level of Performance with respect to Accuracy

Numeric Weight	Verbal Interpretation
3.26 - 4.00	Always Observed
2.51 - 3.25	Observed
1.76 - 2.50	Sometimes Observed
1.00 - 1.75	Not Observed

III. Level of Performance with respect to Accessibility

Numeric Weight	Verbal Interpretation
3.26 - 4.00	Highly Accessible
2.51 - 3.25	Accessible
1.76 - 2.50	Slightly Accessible
1.00 - 1.75	Not Accessible

Data Gathering Procedures

The researchers requested an official permission in conducting the study from the school principal and the Practical Research adviser. Following the approval, a survey validation process was applied to the research instrument. This began to follow up on the survey instrument, which comprised questions regarding the content of ConCalc and the experiences of the respondents with the instrument, from subject matter experts in mathematics education and ICT experts to assess the learning aid's algorithms and evaluate its performance. After that, the participants were given an overview of the study by the researchers, who went over its goal and methodology, especially the survey questionnaire about ConCalc as a virtual tool for mathematics instruction. Students who have given their consent then received the survey questions. Upon collecting the questionnaires, the researchers analyzed the data using appropriate methods of statistics, obeying the guidelines provided in the research plan.

The chosen instrument was used to assess the level of acceptability and the level of performance of Conics Calculator (ConCalc) as educational website for STEM learners.

Statistical Data

This section contains the statistical methods that were utilized in the study to interpret the data gathered from the assessment and questionnaire results.

Variable (s)	Statistical Treatment (s)
The algorithms or equations used in developing ConCalc	
The level of acceptability of ConCalc as educational website	Weighted Mean
The level of performance of ConCalc as educational website	Weighted Mean
Significant relationship between the level of acceptability and the level of performance of ConCalc as educational website	Spearman rho Correlation

Chapter 4

PRESENTATION, ANALYSIS, AND INTERPRETATION OF DATA

This chapter presents the data collected from the research and provides a comprehensive analysis and interpretation of the findings. The data are systematically organized to address the research questions and objectives.

Procedures and Algorithms Used in Developing Conics Calculator (ConCalc) as Educational Website for STEM Learners

The Hypertext Markup Language (HTML), Cascading Style Sheet (CSS), and JavaScript applied in ConCalc

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Home Page</title>
  <link rel="stylesheet" href="style.css">
  <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.2/css/all.min.css">
</head>
<body>

  <header>
    <div class="ceiling">
      <button onclick="sideNavBtn()" class="side-nav-btn"><i class="fa-solid fa-bars-staggered"></i></button>
      <div class="logo">ConCalc</div>
    </div>

    <nav class="nav-bar">
      <ul>
        <li><a onclick="introduction()" href="#home">Home</a></li>
        <li><a onclick="aboutUs()" href="#">About Us</a></li>
      </ul>
    </nav>
  </header>

  <div class="container">
<!--SIDE-BAR-->
    <div class="side-nav">
      <h1 class="side-nav-title">CONICS SECTION</h1>
      <ul>
<!--FOR TRANSFORMING THE CIRCLE-->
        <h1 class="nav-con-title">Circle</h1>
        <div class="transform-btn">
          <button><a href="GenForm - circle/circleGF.html">General - Standard Form</a></button>
          <button><a href="SF - circle/circleSF.html">Standard - General Form</a></button>
        </div>

        <h1 class="nav-con-title">Ellipse</h1>
        <div class="transform-btn">
          <button><a href="ellipse GF/ellipseGF.html">General - Standard Form</a></button>
          <button> <a href="ellipse SF/ellipseSF.html">Standard - General Form</a></button>
        </div>
      </ul>
    </div>
  </div>
</body>
</html>
```

Figure 2

Sample HTML Codes Applied in ConCalc

Figure 2 on the previous page illustrates the Hypertext Markup Language (HTML) codes used in ConCalc to set up the webpage structure, links to external style sheets, employs JavaScript, and incorporates Font Awesome and Google Fonts for custom typography. HTML is the foundation which links all of the content and design elements together in ConCalc. It provides structure to ConCalc by specifying headings, paragraphs, and links.

```

*{
  margin: 0;
  padding: 0;
  font-family: Arial, sans-serif;
}
a{
  text-decoration: none;
}
:root{
  --background-color: hsla(216, 96%, 9%, 0.807);
  --color: #fff;
  --blk-color: #333;
  --header-font: 20px;
  --gray-background: hsla(0, 0%, 50%, 0.17);
}
header{
  display: flex;
  justify-content: space-around;
  align-items: center;
  position: fixed;
  padding: 10px 0;
  width: 100%;
  background-color: var(--blk-color);
}
header .nav-bar li a:hover{
  background: #fff;
  color: #333;
  padding: 3px 8px;
  transition: ease-in 0.3s;
}
header .logo{
  font-size: 30px;
  font-weight: bold;
  margin-left: 200px;
  color: var(--color);
}
header ul{
  display: flex;
}
header ul li{
  list-style: none;
  margin: 0 10px;
}

```

Figure 3

Sample CSS Codes Applied in ConCalc

Figure 3 on the previous page shows the Cascading Style Sheets (CSS) codes used in ConCalc. CSS is a language for styling and designing ConCalc. It works with HTML, which creates the structure of a web page. CSS is used to customize the ConCalc website's colors, fonts, spacing, layout, and other visual components to make it more appealing and user-friendly. It also supports responsive design, which adjusts the website layout to accommodate multiple screen sizes, such as those found on phones and computers.

```
const resultDiv = document.querySelector(".steps-container .circleGF-result");
const form = document.querySelector(".form");
const addContainer = document.querySelector(".add-container");
const step1Container = document.querySelector(".step1");
const step2Container = document.querySelector(".step2");
const step3Container = document.querySelector(".step3");
const AllStep1 = document.querySelector(".step1-container");
const AllStep2 = document.querySelector(".step2-container");
const AllStep3 = document.querySelector(".step3-container");
const AllStep4 = document.querySelector(".step4-container");
const answers = document.querySelector(".circleGF-answer-container");
const graphValue = document.querySelector(".graph-value");
const graph = document.querySelector(".graph");
const userContainer = document.querySelector(".users-answer-container");
const partsContainer = document.querySelector(".parts-container");
const allStep = document.querySelector(".all-steps")

const aValue = document.querySelector(".aValue");
const bValue = document.querySelector(".bValue");
const cValue = document.querySelector(".cValue");
const dValue = document.querySelector(".dValue");
const constValue = document.querySelector(".constValue");
const aDeno = document.querySelector(".aDeno");
const bDeno = document.querySelector(".bDeno");
const cDeno = document.querySelector(".cDeno");
const dDeno = document.querySelector(".dDeno");
const constDeno = document.querySelector(".constDeno");

const sign1 = document.querySelector(".sign1");
const sign2 = document.querySelector(".sign2");
const sign3 = document.querySelector(".sign3");
const centerValue = document.querySelector(".centerValue");
const radiusGraph = document.querySelector(".radius");
const given = document.querySelector(".given");

function showStep1(){
    step1Container.style.display = "block";
    AllStep2.style.display = "block";
    document.querySelector(".step1Btn").style.display = "none";
    document.querySelector(".step2Btn").style.display = "block";
}
```

Figure 4

Sample JavaScript Codes Applied in ConCalc

Figure 4 on the previous page shows the application of JavaScript in ConCalc. It makes use of vanilla JavaScript, which is a way of writing JavaScript code that does not require any other frameworks or libraries. There are no plugins or other libraries present; it is just pure JavaScript code written in its raw form. This programming language makes it possible for ConCalc to have dynamic content, real-time changes, and user interactions.

Level of Acceptability of ConCalc as Educational Website for STEM Learners with respect to Usability and Web Design

Table 1

Level of Acceptability of ConCalc with respect to Usability

Upon testing the ConCalc, the website...	Weighted Mean	Interpretation	Rank
1. guides users through the process and builds mathematical understanding about conversion of forms of conic sections.	3.94	Highly Acceptable	1
2. gives clear and understandable directions for inputting equations.	3.91	Highly Acceptable	2.5
3. presents step-by-step procedures of conversion in an easy-to-follow manner.	3.90	Highly Acceptable	4
4. offers clear and concise lessons or concepts to help students understand conic sections.	3.91	Highly Acceptable	2.5
5. provides users to see graphical representation of each conic sections.	3.89	Highly Acceptable	5
Average Weighted Mean	3.91	Highly Acceptable	

The data from Table 1 reveals excellent results for ConCalc with respect to its usability. The calculated weighted mean reflects positive performance

across several key features including accurate calculations, clear graph displays, and well-organized step-by-step procedures for conic sections, correct equation usage, and precise graphical data representation—all falling within the Highly Acceptable range.

The overall mean of ConCalc with respect to its usability is 3.91, indicating that it is Highly Acceptable. This emphasizes the significance of the website in guiding the users through the process.

To further support this idea, livari (2023) said that websites and applications must now go beyond simple interaction design to offer dynamic, user-driven guidance. The study emphasizes that guiding users involves minimizing confusion and cognitive overload through clear instructions, navigation tools, and proactive support.

Table 2 on the next page shows the results of evaluating the web design of ConCalc, with all aspects rated as Highly Acceptable. The website excels in readability, with easily understandable text, a clean and organized layout, and comfortable font sizes. ConCalc also has clear labeling of buttons and menus, enhancing user navigation. Furthermore, visual content and design elements effectively engage users.

The average weighted mean for web design is 3.82, demonstrating a Highly Acceptable result. It highlights the importance of user-friendly interfaces and responsive design for successful educational technology platforms.

Table 2

Level of Acceptability of ConCalc with respect to Web Design

Upon testing the ConCalc, the website...	Weighted Mean	Interpretation	Rank
1. has an easily readable text in its background.	3.84	Highly Acceptable	2
2. has a clean and organized layout that improves readability and focus of the users.	3.89	Highly Acceptable	1
3. has clearly labeled buttons and menus.	3.83	Highly Acceptable	3
4. enhances engagement of the users with its visual content and design.	3.71	Highly Acceptable	5
5. has fonts and text size that are comfortable to read.	3.81	Highly Acceptable	4
Average Weighted Mean	3.82	Highly Acceptable	

Huth et al. (2024) said that visual enhancements in text affect reading behavior. It finds that visualization techniques, such as highlights and icons, can improve reading behavior, suggesting that well-designed educational content can enhance readability and user focus.

Level of Performance of ConCalc as Educational Website for STEM Learners with respect to Accuracy and Accessibility

Table 3 on the next page assesses the performance of ConCalc as an educational website for STEM learners, focusing on accuracy. ConCalc receives Always Observed ratings for its accuracy, demonstrating strong performance in providing accurate calculations, displaying graphs clearly, presenting well-organized step-by-step procedures, showing correct equations, and supplying precise values on graphs.

Table 3

Level of Performance of ConCalc with respect to Accuracy

Upon testing the ConCalc, the website...	Weighted Mean	Interpretation	Rank
1. gives an accurate calculations and results.	3.91	Always Observed	1.5
2. provides correctly and clearly displayed graph.	3.90	Always Observed	3.5
3. presents correct and well-organized step-by-step procedures in converting forms of conics section.	3.90	Always Observed	3.5
4. shows correct equation used for calculations and produces expected results.	3.91	Always Observed	1.5
5. supplies precise values represented on graphs based on the input data.	3.87	Always Observed	5
Average Weighted Mean	3.90	Always Observed	

The overall mean score of 3.90 indicates a high level of accuracy. This aligns with research highlighting the importance of accurate and reliable tools in STEM education. It is essential for educational platforms to not only offer accurate calculations but also to provide expected results.

According to the book of Ortiz (2020), educational technologies such as calculators, software tools, and apps, provide precise and accurate results, ensuring students can rely on the technology to verify their solutions and also not only deliver correct results but also help students visualize the underlying processes.

Table 4 on the next page shows the accessibility of ConCalc for STEM learners which is rated as Highly Accessible. The website receives positive rating for ease of navigation of information, usability on small screen devices, providing

key shortcuts for easier navigation, content suitable for STEM learners, and loading quickly and providing quick answers.

Table 4

Level of Performance of ConCalc with respect to Accessibility			
Upon testing the ConCalc, the website...	Weighted Mean	Interpretation	Rank
1. navigates and finds information that the STEM learners need.	3.90	Highly Accessible	1
2. can be used even in small screen devices.	3.83	Highly Accessible	3
3. provides key shortcuts for easier navigation.	3.74	Highly Accessible	5
4. has a content suitable for the STEM learners, avoiding overly complex language.	3.87	Highly Accessible	2
5. loads and provides quick answer.	3.79	Highly Accessible	4
Average Weighted Mean	3.83	Highly Accessible	

The overall mean score for accessibility was 3.83 rated as Highly Accessible. These results show that the website helps the user to navigate information needed.

Based on the study of Lazar et al. (2015), it highlights the significance of accessible navigation systems for users with disabilities. This suggests that poor navigation systems can significantly hinder information retrieval for visually impaired users, as they often rely on keyboard navigation and screen readers.

Relationship between the Level of Acceptability of ConCalc and its Level of Performance

The analysis in Table 5 on the next page reveals a statistically significant relationship between the usability, accuracy, and accessibility of the website. The

computed values of 7.59 and 5.71 exceed the tabular value of 1.995, which indicates that the null hypothesis was rejected. In both cases, the correlational values of 0.68 and 0.57 indicate a moderately strong positive correlation.

Table 5

**Relationship between the Level of Acceptability of ConCalc
with respect to Usability and its Level of Performance
with respect to Accuracy and Accessibility**

Variables	Statistical Treatment	Degree of Freedom	Correlational Value	Computed Value	Tabular Value	Decision	Interpretation
Usability Accuracy	Spearman's rho	68	0.68	7.59	1.995	Reject H_0	Significant
Usability Accessibility	Spearman's rho	68	0.57	5.71	1.995	Reject H_0	Significant

The results suggest that as the level of acceptability with respect to usability increases, the level of performance with respect to accuracy and accessibility also tend to increase and vice versa. It implies that the ConCalc guides the users and builds mathematical understanding. It also gives accurate results, shows correct equations used for calculations, and navigates and finds information for STEM learners.

In contrast, according to a systematic review by Molina et al. (2022) which focused on the mathematics instructions, it found that usability assessments often emphasize criteria like efficiency and ease of use, sometimes neglecting accessibility considerations. This suggests that enhancing usability and efficiency might negligently compromise accessibility for some users. Additionally, based on Anonymous (2015), online tools and websites often fall short in developing deep conceptual understanding, particularly in mathematics. The study points out that while websites may present accurate calculations, they often do not provide

sufficient explanation or contextualization of the steps involved. This can lead to a "procedural" understanding rather than a "conceptual" understanding of math.

Table 6

**Relationship between the Level of Acceptability of ConCalc
with respect to Web Design and its Level of Performance
with respect to Accuracy and Accessibility**

Variables	Statistical Treatment	Degree of Freedom	Correlational Value	Computed Value	Tabular Value	Decision	Interpretation
Web Design Accuracy	Spearman's rho	68	0.54	5.31	1.995	Reject H_0	Significant
Web Design Accessibility	Spearman's rho	68	0.53	5.17	1.995	Reject H_0	Significant

Table 6 shows the results of the statistically significant relationship between the web design, accuracy, and accessibility of the website. Since the computed values of 5.31 and 5.17 exceed the tabular value of 1.995, the researchers reject the null hypothesis, indicating a significant correlation. The correlational values of 0.54 and 0.53 show moderately strong positive correlation.

This suggests that as the level of acceptability with respect to web design increases, the level of performance with respect to accuracy and accessibility will also tend to increase and vice versa. This implies that ConCalc has a clean and organized layout.

This negates the study of Macgilchrist et al. (2024), revealing that accuracy in educational technology is influenced more by contextual implementation and content integrity than by aesthetics or interface design of the website. Furthermore, based on the journal of Shaheen et al. (2017), it reveals that by analyzing accessibility issues within K–12 educational settings, the research underscores the critical role of web design in ensuring inclusivity. The

findings demonstrate that poorly designed websites exclude users who rely on assistive technologies, contradicting the notion that web design has minimal impact on accessibility.

Chapter 5

SUMMARY OF FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS

This chapter summarizes the key findings of the study, presents the conclusions based on the results, and offers recommendations for future action.

Summary of Findings

Based on the thorough study, the following findings are hereby summarized:

1. Procedures and Algorithms Used in Developing Conics Calculator (ConCalc) as Educational Website for STEM Learners

To further discuss the website, the researchers provided a booklet containing codes and algorithms. They created educational materials to help STEM students deal with conic sections by integrating these complex equations. The booklet serves as a useful resource for scholars, assisting them in developing their own instructional resources and improving the field of education technology.

2. Level of Acceptability of ConCalc as Educational Website for STEM Learners with respect to Usability and Web Design

2.1 With respect to Usability, ConCalc was interpreted as Highly Acceptable with an average weighted mean of 3.91.

2.2 With respect to Web Design, ConCalc was interpreted as Highly Acceptable with an average weighted mean of 3.82.

3. Level of Performance of ConCalc as Educational Website for STEM Learners with respect to Accuracy and Accessibility

3.1 With respect to Accuracy, ConCalc was interpreted as Always Observed with an average weighted mean of 3.90.

3.2 With respect to Accessibility, ConCalc was interpreted as Highly Accessible with an average weighted mean of 3.83.

4. Relationship between the Level of Acceptability of ConCalc and its Level of Performance

Spearman's rho correlation was used in the study to examine the significant relationship between the following variables: usability, web design, accuracy, and accessibility. The results presented that the computed values continue to exceed the tabular value, having significant correlations between the corresponding pairs of variables. The overall relationship of the variables reveals moderately strong positive correlation.

Conclusions

The findings of the research led to the following conclusions:

1. The functionality of ConCalc, which includes well-structured algorithms and clear explanations, contributes to its usability, web design, accuracy, and accessibility as an electronic learning resource.
2. ConCalc is a valuable educational tool for STEM learners in dealing with conic sections, particularly regarding the tool's usability and web design.
3. ConCalc is an effective tool for helping STEM learners to have accurate answers and accessible learning material.

4. ConCalc guides the users and builds mathematical understanding. It has a clean and organized layout, gives accurate results, shows correct equations used for calculations, and navigates and finds information for STEM learners.

Recommendations

Based on the conclusions drawn, the researchers recommend the following:

1. The researchers recommend integrating ConCalc in Pre-Calculus subject as an instructional material in STEM classes.
2. Future researchers are recommended to explore the impact of ConCalc on student learning outcomes (e.g., improved test scores, increased confidence) through experimental studies.
3. The researchers also recommend improving the visual content and design of the ConCalc by adding vibrant colors and improving the image quality of the graph.
4. The researchers propose adding a factoring calculator as an additional feature of the website.
5. The researchers suggest adding an exercise quiz feature to improve student engagement.

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